Hierarchical Regionalization for Connected Redistricting Construction

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Draft — May 12, 2026

Abstract

This draft presents T.16, a hierarchical regionalization baseline for BISECT. The paper focuses on deterministic agglomeration, merge witnesses, capacity and contiguity checks, and RPLAN-compatible lineage. Regionalization is framed as a construction family that makes the sequence of region merges inspectable, not as proof that the merge sequence is optimal, legally sufficient, or politically fair.

1 Introduction

Regionalization methods construct plans by merging small connected units into larger connected regions. T.16 adds this family to BISECT so the merge sequence itself becomes part of the explanation and audit trail.

2 Method

The staged baseline performs deterministic agglomerative merging while tracking merge witnesses. Each witness records enough information to explain why regions were combined and to distinguish valid construction from infeasible capacity cases.

The atlas reading rule is that larger regions are not self-explanatory. The paper must show the adjacent-pair candidate set, the eligibility gate, the score used only after eligibility is established, and the merge witness that creates the next non-leaf region. Rejected adjacent pairs remain evidence when they explain why an apparently plausible merge was not taken.

2.1 Reproducible Procedure

The constructor operates on the adjacency graph emitted by the BISECT state pipeline. Connectedness is defined over that graph: candidate regions and final districts must be connected under the declared profile. Islands, disconnected components, and alternative adjacency definitions are therefore input-profile questions rather than claims solved by the merge routine.

1. Initialize each unit as a singleton region with population and adjacency metadata.

2. Enumerate adjacent candidate region pairs that remain eligible under the declared capacity and connectedness profile.
3. Score candidate merges using the implementation's deterministic merge scoring rule.
4. Select the best candidate, breaking ties by stable region identifiers.
5. Record a merge witness and repeat until the requested number of regions is reached or no eligible merge remains.

Witness field	Purpose
Input region ids	Identifies the merged regions
Candidate score	Explains merge ordering
Constraint status	Records capacity/connectedness checks
Result region id	Links the new region to descendants
Parameter hash	Binds merge settings to the artifact

Table 1: Minimum merge-witness fields for replayable regionalization lineage.

3 Implementation

The implementation lives in `bisect-clustering::regionalization` and is exposed through `bisect state --structure regionalization`. Outputs include regionalization summaries, merge logs, parameter hashes, and algorithm lineage.

4 Audit Artifacts

Regionalization audit artifacts record the final assignments together with merge witnesses, parameter hashes, and algorithm lineage in the shared RPLAN/RCTX sidecar flow. The fixed-point claim is reproducibility and declared profile verification, not proof that a merge sequence is uniquely correct or legally sufficient.

The verifier can check that the exported plan and sidecars agree with the declared profile and recorded lineage. It does not certify that the merge order is globally optimal, that communities were preserved, or that the plan satisfies legal standards beyond the explicit audit profile.

5 Evaluation Plan

The current evidence layer includes path, grid, two-clique, capacity, and determinism fixtures, plus a benchmark-tier RPLAN package generated from the regionalization constructor on a 100-unit synthetic path. The next evidence layer should compare regionalization against flat clustering, METIS, and flow construction on selected real-data smoke states.

Claim	Evidence now	Missing evidence	Status
Connected fixture output	L0 path/grid tests	None for fixtures	Current
Deterministic merge logs	L0 determinism tests	Broader graph sweep	Current
Capacity handling	L0 capacity fixtures	Real-data smoke	Current
Audit lineage	L1 sidecar checks	Real export transcript	Current
Scale smoke	path100 benchmark package	Real-data benchmarks	Benchmark smoke
Quality vs. baselines	None in this paper	Commands and outputs	Future

Table 2: Evidence ladder for T.16 claims. Connectedness, lineage, and empirical quality are separate evidence levels.

Constructor	Primary witness	Failure artifact	Current role
Regionalization	merge witness	no eligible merge/status	baseline
Capacity clustering	repair/status	infeasibility witness	comparison
Flow construction	flow summary	infeasibility witness	comparison
Spectral	split lineage	validation failure	comparison
METIS	partition manifest	validation result	comparison

Table 3: Failure-behavior comparison framing. Quantitative comparisons remain future work until commands and outputs are recorded.

6 Limitations

The current implementation is a deterministic regionalization baseline, not a complete SKATER or Max-p reproduction. Merge scoring choices can affect output quality and should be evaluated empirically before broad claims.

Merge lineage improves inspectability but does not make the merge sequence normatively correct. Connectedness and capacity checks do not imply compactness, community preservation, partisan fairness, or legal compliance beyond the declared audit profile.

7 Conclusion

T.16 makes hierarchical regionalization a first-class construction path in BISECT and preserves the same audit contract used by other final-plan producers. The contribution is replayable merge provenance under a declared profile, with empirical district-quality claims reserved for future recorded sweeps.

References

BISECT Project. T.16 hierarchical regionalization implementation spec, 2026. BISECT internal specification.